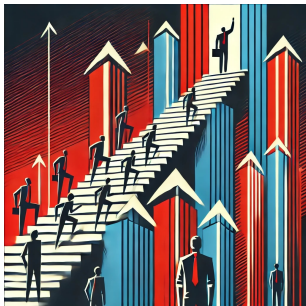


PROMOTIONS AND SELECTION



INCENTIVES SO FAR

Performance pay induces effort, but

- ▶ Can be expensive if agents are risk averse
- ▶ May cause effort diversion if multitasking
- ▶ Crowd out intrinsic motivation

Today:

1. Holistic evaluations, Yardstick competition and Promotions
2. Selection and Hiring

OUTLINE

HOLISTIC BUT SUBJECTIVE EVALUATIONS

YARDSTICK COMPETITION AND PROMOTIONS

SELECTION

KEY TAKEAWAYS

HOW TO SOLVE MULTITASKING?

Multitasking:

- ▶ Performance on some tasks that agent performs are difficult to measure
- ▶ Incentivizing those tasks that are easier to measure leads to effort distortion
- ▶ **Solution:** Find something that measures effort on all tasks

A possibility: Holistic evaluation by supervisor

HOLISTIC EVALUATIONS

Principal may know how well the Agent performed

- ▶ Principal knows who is a good teacher
- ▶ Principal may have a good sense if a manager takes steps to prevent fraud but also works hard to uncover it

Jacob and Lefgren (JLE 2008) compared several possible predictors of teacher performance

- ▶ Experience
- ▶ Education
- ▶ Past value-added
- ▶ Principal's rating

Principal's rating is best predictor of future student outcomes

PROBLEMS WITH HOLISTIC EVALUATIONS

Using **holistic evaluation** to reward the agent may allow to provide incentives without causing multitasking problem

Holistic evaluations by a supervisor are **subjective**

- ▶ They cannot be verified by court

The principal faces a **commitment** problem

- ▶ Wants to promise to pay bonus if agent performs well
- ▶ Ex post has incentive to renege on her promise

How can principal solve this problem?

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RELATIVE PERFORMANCE

Typically a principal hires more than two agents

Using **relative performance** may help with multitasking and risk

YARDSTICK COMPETITION

An investment firm employs two financial consultants managing client portfolios in different regional markets

Consider two incentive schemes:

- ▶ **Performance pay:** pay each consultant based on the absolute return generated for clients
- ▶ **Yardstick competition:** pay based on how **much higher / lower** the consultant's return is compared to the other consultant

Which approach is better?

FRAMEWORK FOR YARDSTICK COMPETITION

Portfolio performance is determined by:

- ▶ Consultant's effort and skill
- ▶ Luck, which may have two components
 - ▶ A **common shock**: affects both consultants (e.g., macroeconomic downturn, interest rate changes, market volatility)
 - ▶ An **idiosyncratic shock**: affects each consultant differently (e.g., regional economic conditions, sector exposure, client risk profiles)

FRAMEWORK FOR YARDSTICK COMPETITION

► $y_1 = e_1 + luck_{common} + luck_1$

► $y_2 = e_2 + luck_{common} + luck_2$

► Performance pay:

$$by_1 = b(e_1 + luck_{common} + luck_1)$$

$$var(w) = b^2(var(luck_{common}) + var(luck_1))$$

► Yardstick competition:

$$b(y_1 - y_2) = b(e_1 - e_2 + luck_1 - luck_2)$$

$$var(w) = b^2(var(luck_1) + var(luck_2))$$

If common shock plays a big role in shaping performance, tying bonuses to it is very risky for agents

Yardstick competition removes this risk

WHEN IS YARDSTICK COMPETITION A GOOD IDEA FOR PRINCIPAL?

Yardstick competition

- ▶ Performs **as well as** standard performance pay if agents are risk neutral
- ▶ Performs **better** if agents are risk averse, output is affected by large **common** shock and **idiosyncratic** shocks are small
- ▶ Performs **worse** if agents are risk averse, output is affected by small **common** shock and **idiosyncratic** shocks are large

PROMOTIONS

Promotions (Tournaments): yardstick competition in which a fixed reward is given to only the top performer(s)

- ▶ Agents at same hierarchy level compete for promotion
- ▶ Promotion is valuable
 - ▶ Higher salary
 - ▶ Higher prestige
- ▶ Winner is based on relative performance
- ▶ Common shock does not affect who gets promoted

Since promotions are meant to incentivize effort, they often come with big salary increases, even if job is not much harder

YARDSTICK COMPETITION AND PROMOTIONS AS SOLUTIONS TO COMMITMENT PROBLEM WITH SUBJECTIVE EVALUATIONS

Suppose you want to use subjective evaluations

Commit to the promotion schedule:

- ▶ The number of people who are promoted each year
- ▶ The size of salary increase with promotion

Principal is committed to promotion, so she may as well promote those who perform best

Commit to the bonus pool

- ▶ Those performing better receive higher bonus

Principal is committed to bonus pool, so she may as well reward those who perform best

EXAMPLES: SALES

Sales teams frequently use “relative commission”

A sales organization sets a bonus pool, then pays more to reps who outperform other reps

Similar arrangement are used by call centers

Main reason: remove common shock

EXAMPLE: GOVERNMENT BUREAUCRACY

Government jobs frequently require multiple tasks and performance is hard to measure

Governments frequently rely on promotions based on supervisors' evaluations or yardstick competition with fixed bonus pool:

- ▶ U.S. Federal Government: Senior Executive Service (SES) bonus pools (explicit “tournament inside the agency”)
- ▶ UK Civil Service: end-of-year performance-related pay comes from a constrained pool

ISSUES WITH YARDSTICK AND PROMOTIONS

1. Agents could collude and all exert low effort
2. Agents could sabotage one another
3. May create hostile environment
4. Discouragement effect:
 - ▶ If I know I am behind, why bother?
5. If based on subjective evaluation, then susceptible to bias and favoritism



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KEY TAKEAWAYS

BEFORE WE HIRE THE WORKER...

So far, we talked about how to incentivize the workers we have....

...we should also hire the right workers!

Some characteristics of candidates may be **observed** (easy problem) while others are **not easily observed** (harder problem)

CHARACTERISTICS ORGANIZATION MAY VALUE

1. Intrinsically motivated
2. Skilled (meaning: effort translates in performance)
3. Committed to stay on the job for a long time
4. Aligned with the principal
5. Well-informed about the task (next lecture)

Principal chooses best based on observable characteristics

What can she do if she can't tell which agent is which?

Incentive schemes affect the kind of workers you attract

ADDRESSING 1: INTRINSIC MOTIVATION

In your homework!

ADDRESSING 2: HIGH-POWERED INCENTIVES ATTRACT SKILLED AGENTS

Consider two classes of risk-neutral agents

- ▶ Skilled: effort affects output: $y_C = e + luck$
- ▶ Unskilled: output depends entirely of luck: $y_I = luck$

Offer bonus $b^* > \alpha$ and fixed salary $s^* = u - (b^* - \alpha) < u$

Only skilled agents take the contract!

By lowering base salary and compensating with higher bonus,
we attract only skilled agents

EXAMPLE OF INCENTIVES SELECTING SKILLED AGENTS: TEACHERS

There are two kinds of teachers:

- ▶ High ability—raise test scores if they work hard
- ▶ Low ability—bad test scores no matter what

What happens if some school districts pay incentive pay and others do not?

DEE AND WYCKOFF 2014

Investigate the effects of IMPACT teacher evaluation program

- ▶ Holistic Measure: Observation and Test Scores
- ▶ Incentive pay and dismissal threat

Results:

- ▶ Voluntary attrition of low performing teachers
- ▶ Improved retention of high performing teachers
- ▶ Improved performance of teachers, in particular high quality teachers

ADDRESSING 3: PROMOTIONS SELECT COMMITTED WORKERS

Consider two classes of agents:

- ▶ Committed—plan to stay at the organization a long time
- ▶ Uncommitted —likely to quit after a short time

The organization wants to hire a committed one:

- ▶ Workers learn on the job and become more productive over time
- ▶ The hiring process is often costly and plagued with uncertainty

How can it avoid hiring uncommitted workers?

ADDRESSING 3: PROMOTIONS

One approach:

- ▶ Offer low initial pay
- ▶ Followed by a steep reward system (e.g., bonuses and promotions after several years of work)

Committed types don't mind waiting for future payments

Uncommitted types do not like this plan—they don't expect to be around long enough to benefit

Promotions with low initial pay screen out uncommitted types

HIRING AI DEVELOPERS

There is fierce competition between companies working on AI to hire the most talented people

Some tech companies are mature, such as Google and Meta

Those companies cannot easily offer equity to the new employees: **equity is a performance pay with steep bonus**

Startup companies such as Deepseek, OpenAI, and ThinkingMachinesLab can easily offer equity to new employees

How do you expect that this affects the type of workers that the companies can attract?

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KEY TAKEAWAYS

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Promotions and Yardstick competition:

- ▶ Remove common luck component
- ▶ Allow to use holistic but subjective evaluations
- ▶ Attract committed agents

Performance pay

- ▶ Increasing bonuses and lowering baseline pay may attract skilled workers
- ▶ Carefully tailoring bonuses and baseline wage may help select intrinsically motivated workers